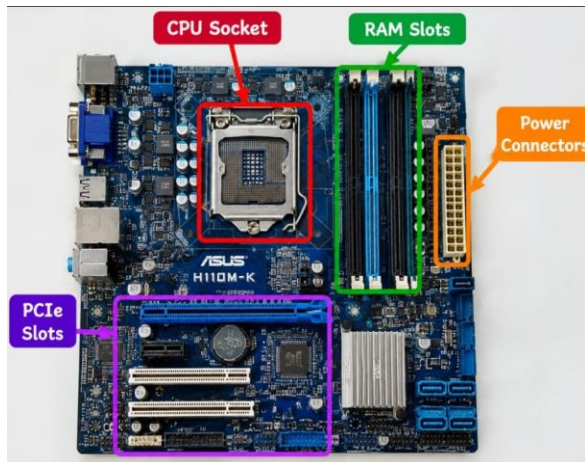


TASK

Question:1

Take a clear photo of the motherboard inside your own or a friend's laptop/desktop (or find a labeled image online), and label the CPU socket, RAM slots, PCIe slots, and PCLE slots, and power connectors using any image editing tool ?

Answer:



Motherboard and CPU Research Assignment

1. Motherboard Image and Labels

Use a clear motherboard image and label these four parts:

Labels to put on the image:

1. CPU Socket

The CPU socket is the place where the processor is installed.

2. RAM Slots / DIMM Slots

These slots are used to install RAM modules.

3. PCIe Slots

PCIe slots are used for expansion cards such as graphics cards, Wi-Fi cards, and sound cards.

4. Power Connectors

Power connectors supply electricity to the motherboard and CPU. Common connectors are the 24-pin ATX connector and 8-pin CPU power connector.

Question:2

Pick any three popular CPUs used in gaming laptops or desktops (for example, Intel Core i5-12400F, AMD Ryzen 5 5600, Intel Core i7-13700K). Research and create a table comparing their clock speed (GHz), number of cores/threads, cache size (L2/L3), TDP (W), -- current price in India?

Specification	i5-12400F	Ryzen5 5600	i7-14700k
Clock speed	2.5-4.4 GHz	3.5-4.4 GHz	3.4-5.4 GHz
Cores/Threads	6/12	6/24	16/24
L2 Cache	7.5 MB	3 MB	24 MB
L3 Cache	18 MB	32 MB	30 MB
TDP	65 W	65 W	125 W
Price in India	₹15,500	₹13,131	₹33,985

Question:3

For each CPU in your comparison table, identify at least one compatible motherboard model (mention brand and model, e.g., ASUS PRIME B660M-A for Intel 12th Gen) and specify the socket type and form factor (ATX, MITx, ITX)?

Answer: 1. Intel Core i5-12400F Compatible Motherboard: ASUS PRIME B660M-A D4

Socket Type: LGA 1700 Form Factor: Micro-ATX ASUS Global

2.AMD Ryzen 5 5600 Compatible Motherboard: MSI B550-A PRO Socket Type: AM4 Form Factor MSI India +1

3.Intel Core i7-13700K Compatible Motherboard: ASUS TUF Gaming Z790-PLUS WIFI D4 Socket Type: LGA 1700 Form Factor

Question:4

Visit the official website of any one CPU you compare (Intel or AMD) and find the maximum supported RAM speed and PCIe version for that CPU. Write these details along with how they affect real-world performance in gaming or multitasking.

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Hint: Look for 'Memory Types' and 'PCI Express Revision' in the CPU specs page?*

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Answer: CPU Selected: Intel Core i5-12400

I checked the official Intel specifications for the Intel Core i5-12400 processor.

1. Maximum Supported RAM Speed:

DDR5: 4800 MT/s

DDR4: 3200 MT/s

2. PCI Express Version:

PCIe 5.0 and PCIe 4.0

Up to 20 PCIe lanes are supported.

3. Effect on Real-World Performance:

. Faster RAM can help the CPU access data quickly and improve multitasking and overall system responsiveness.

. PCIe 5.0 provides higher data transfer bandwidth for compatible devices such as GPUs and SSDs.

. In gaming, the CPU, RAM, GPU and storage work together, so faster supported components can help reduce data-transfer limitations.